

Prove the following are true of all integers:

a) $\text{even} - \text{even} = \text{even}$

b) $\text{odd} - \text{odd} = \text{even}$

c) $\text{even} - \text{odd} = \text{odd}$

d) $\text{odd} - \text{even} = \text{odd}$

Proof of a

Let $\gamma, \phi \in \mathbb{Z}^{\text{even}}$, then

$$\gamma = 2\mu \text{ and } \phi = 2\varepsilon, \text{ and we have}$$

$$\gamma - \phi = 2\mu - 2\varepsilon = 2(\mu - \varepsilon)$$

$$\mu - \varepsilon \in \mathbb{Z}$$

∴ $\gamma - \phi$ is an even integer and we conclude
that the Proposition was correct.

Proof of b:

Let $\gamma, \phi \in \mathbb{Z}^{\text{odd}}$, then

$$\gamma = 2\mu + 1 \text{ for some } \mu \in \mathbb{Z}$$

$$\phi = 2\varepsilon + 1 \text{ for some } \varepsilon \in \mathbb{Z}$$

Then

$$\gamma - \phi = (2\mu + 1) - (2\varepsilon + 1) = 2\mu - 2\varepsilon = 2(\mu - \varepsilon)$$

$$\text{we know } \mu - \varepsilon \in \mathbb{Z}$$

∴ $\gamma - \phi$ is an even integer, and we conclude

that our Proposition was correct.

Proof of c:

Let $\gamma \in \mathbb{Z}^{\text{even}}$ and $\phi \in \mathbb{Z}^{\text{odd}}$, then

$$\gamma = 2\mu \text{ for some } \mu \in \mathbb{Z}$$

$$\phi = 2\varepsilon + 1 \text{ for some } \varepsilon \in \mathbb{Z}$$

Then

$$\gamma - \phi = 2\mu - 2\varepsilon + 1 = 2(\mu - \varepsilon) + 1$$

$$\text{we know } \mu - \varepsilon \in \mathbb{Z}$$

∴ $\gamma - \phi$ is an odd integer and we conclude that

our Proposition was correct.

Proof of d:

Let $\gamma \in \mathbb{Z}^{\text{odd}}$ and $\phi \in \mathbb{Z}^{\text{even}}$, then

$$\gamma = 2\mu + 1 \text{ for some } \mu \in \mathbb{Z}$$

$$\phi = 2\varepsilon \text{ for some } \varepsilon \in \mathbb{Z}$$

Then

$$\gamma - \phi = 2\mu + 1 - 2\varepsilon = 2(\mu - \varepsilon) + 1$$

We know that $\mu - \varepsilon \in \mathbb{Z}$

∴ $\gamma - \phi$ is odd and we conclude that our Proposition is correct.